

TestProject — Project Report

Whether dilps block crowding effects on lifespan.

Experiment type:	Standard Lifespan
Members:	2
Generated:	2026-08-27 21:38
Path:	/home/scott/GithubLocal/pySurvAnalysis/TestProject

2 of 2 member experiment(s) analysed.

Members

Member inventory

Member	Analysed	N	Deaths	Censored	Treatments	Factors	Exclusion group
DataSet1	2026-08-23T12:43:26	5600	777	4823 (86.1%)	4	Sex, Density	none
DataSet2	2026-08-27T21:38:44	5600	5402	198 (3.5%)	4	Sex, Density	none

Each member is analysed independently; nothing here is pooled.

No divergence detected between members: they declare the same factors, levels, censoring policy and exclusion group.

DataSet1

Standard Lifespan

Experiment summary

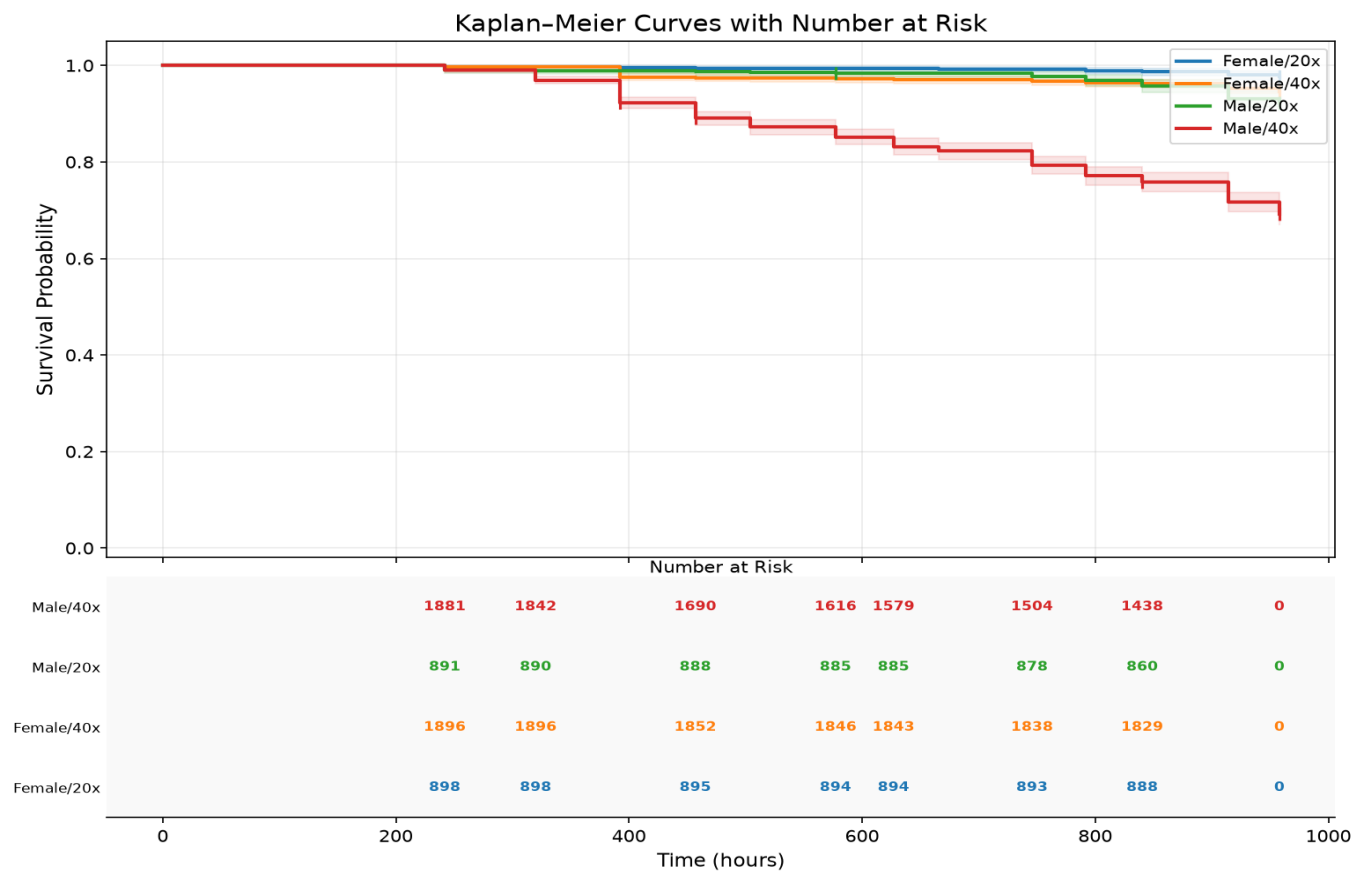
Per-treatment summary

treatment	n_individuals	n_deaths	n_censored	pct_censored
Female/20x	900	20	880	97.800
Female/40x	1900	102	1798	94.600
Male/20x	900	70	830	92.200
Male/40x	1900	585	1315	69.200

Observation window 241.94–958.05, 200 chamber(s).

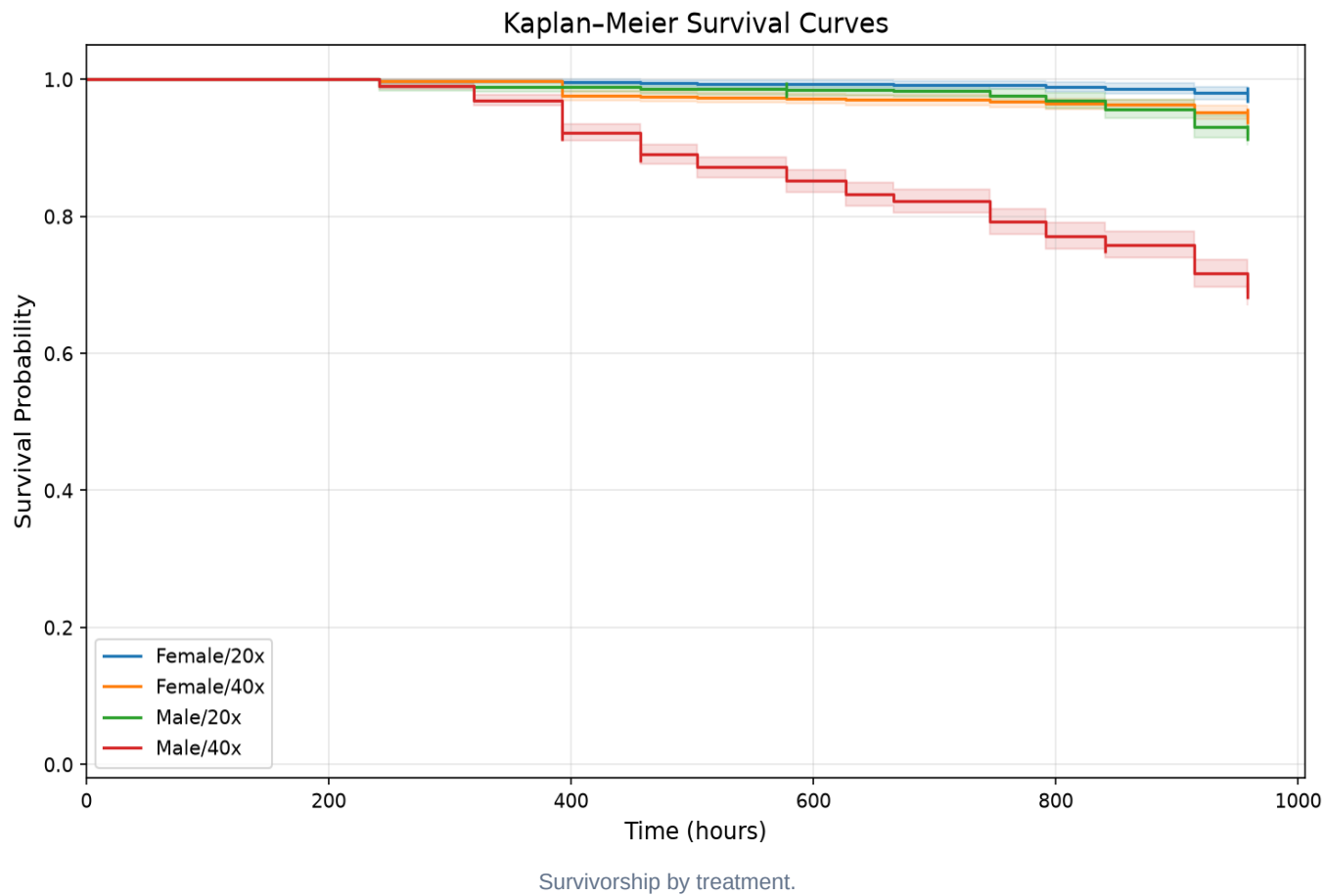
Survivorship figures

KM curves with at-risk table — headline figure

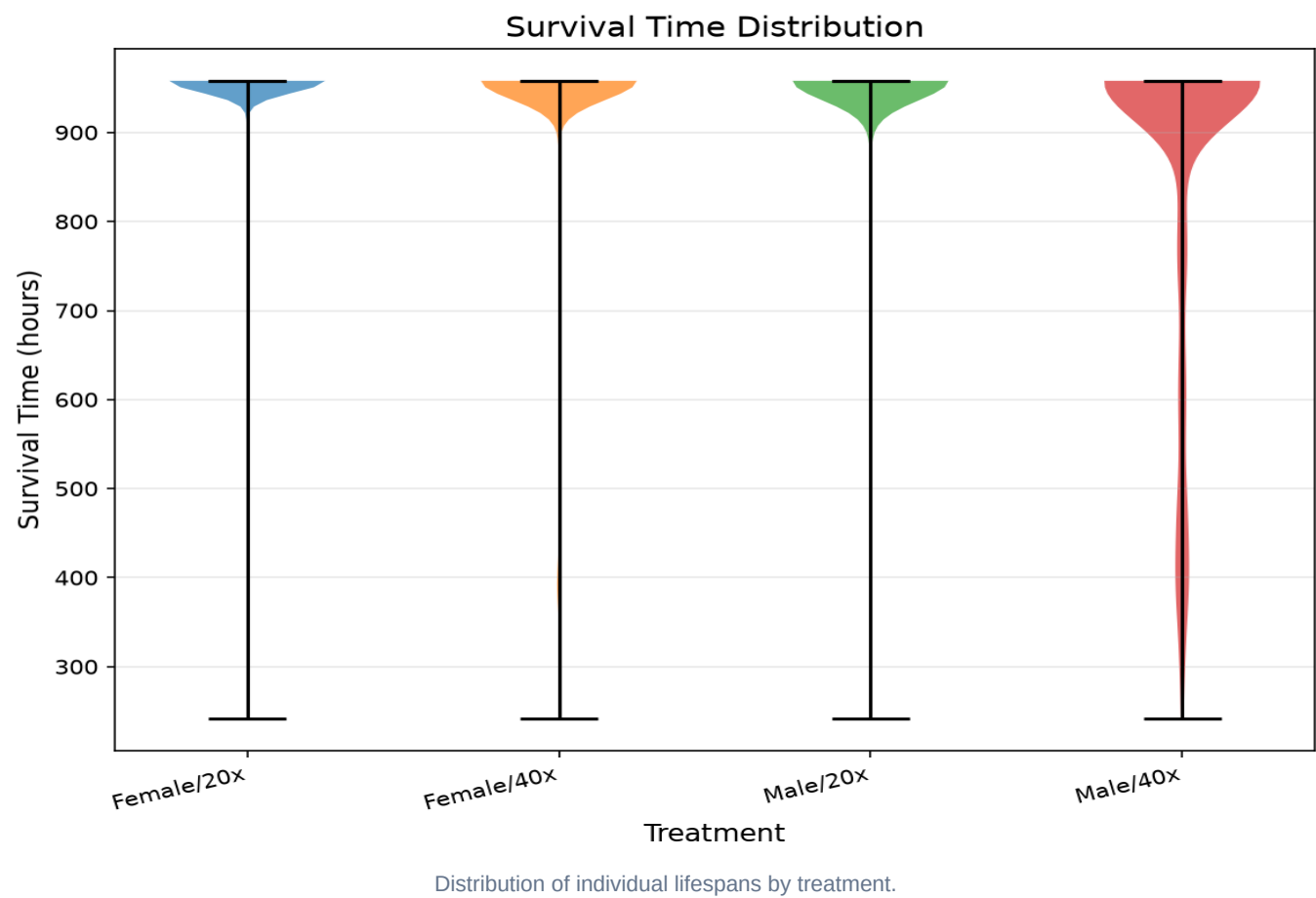


Survivorship with the number at risk beneath the axis.

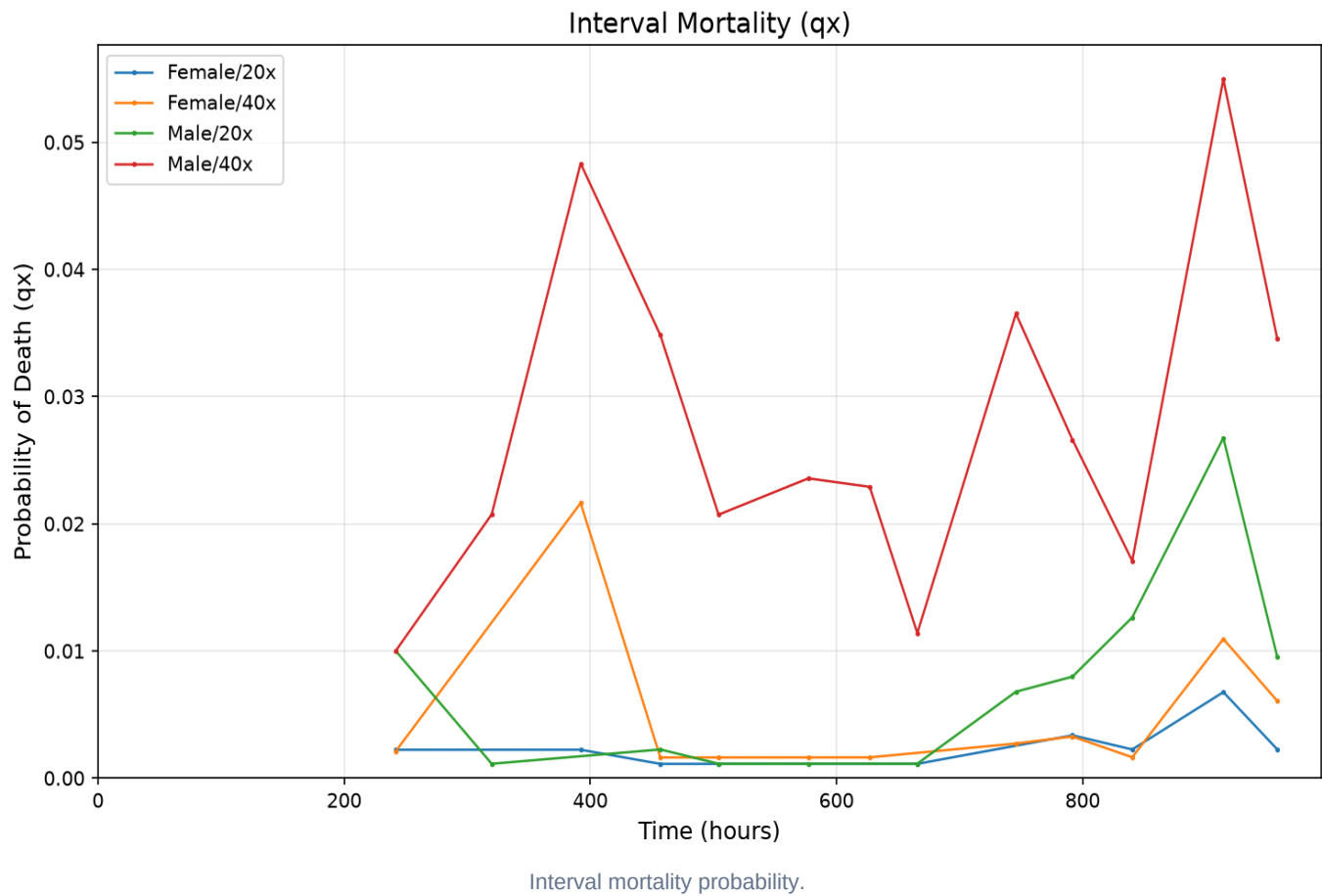
Kaplan-Meier curves



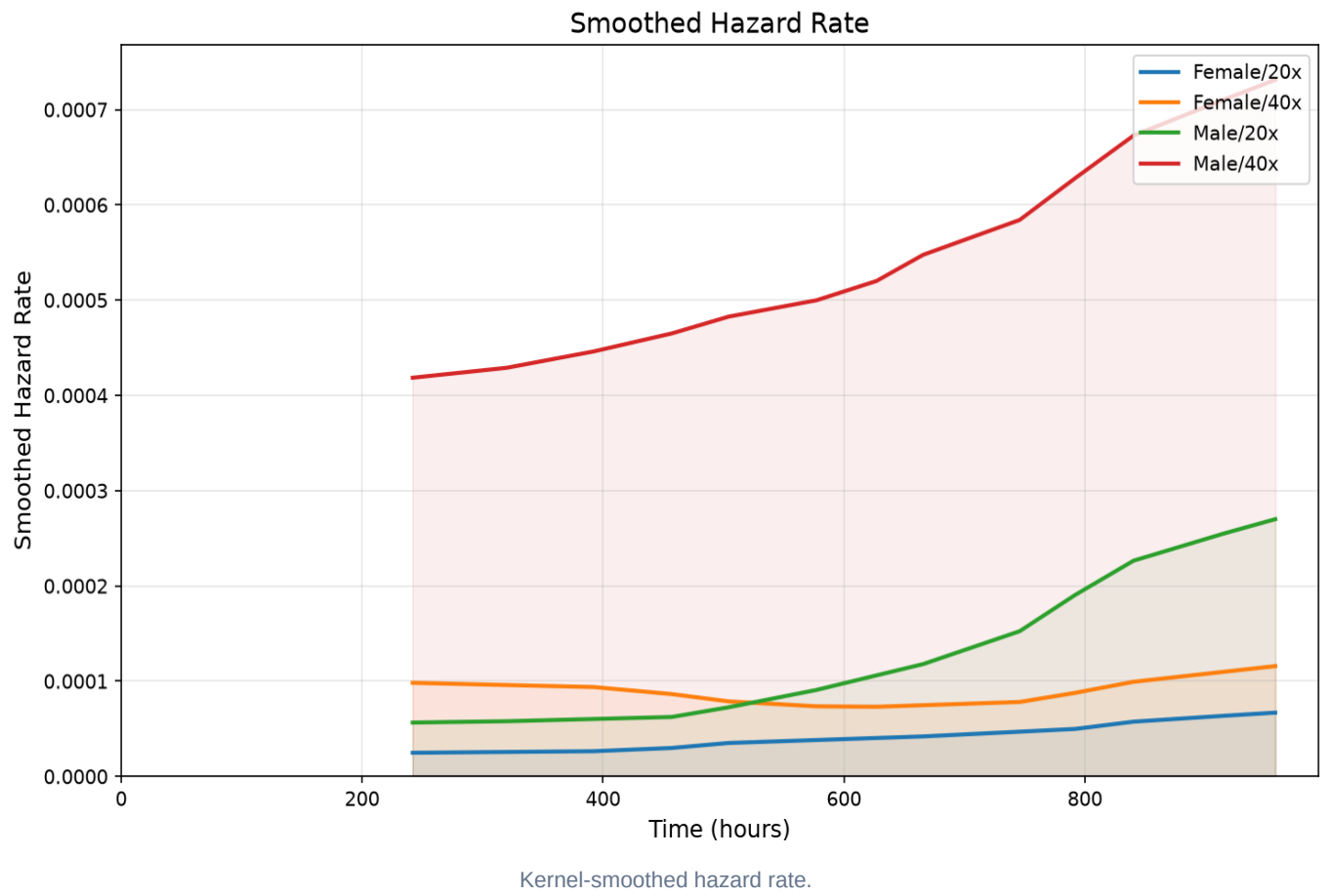
Lifespan distribution



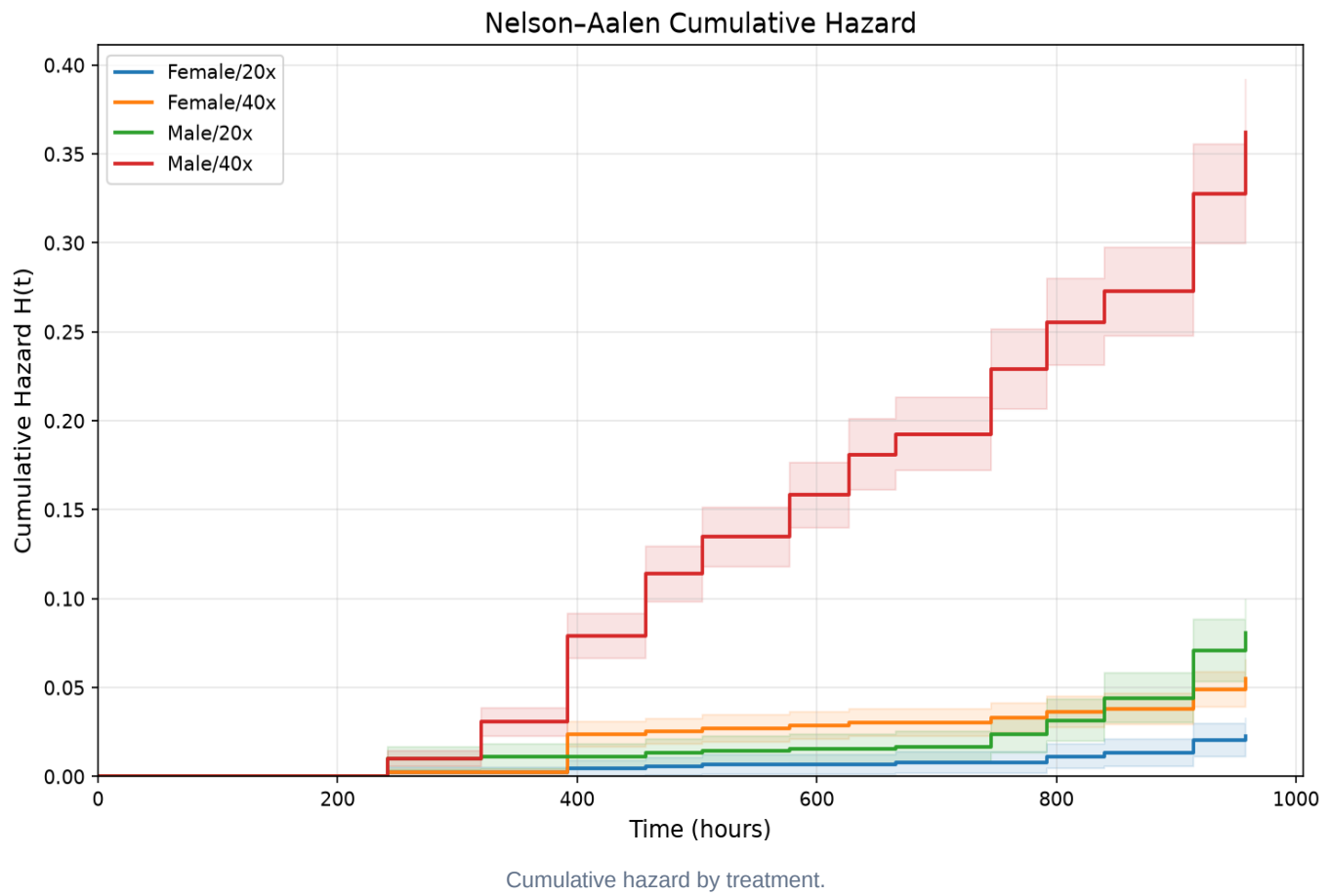
Mortality (qx)



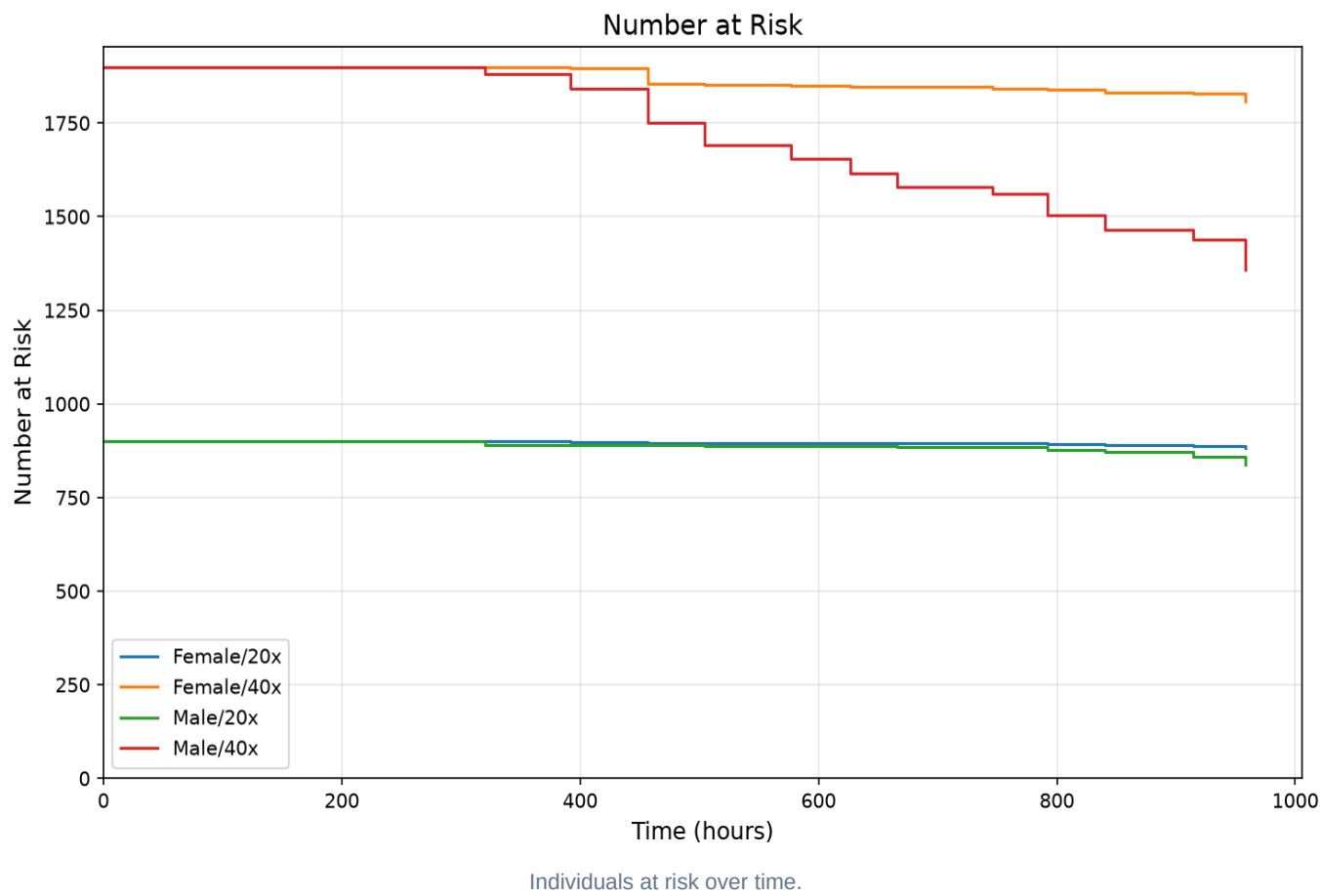
Smoothed hazard



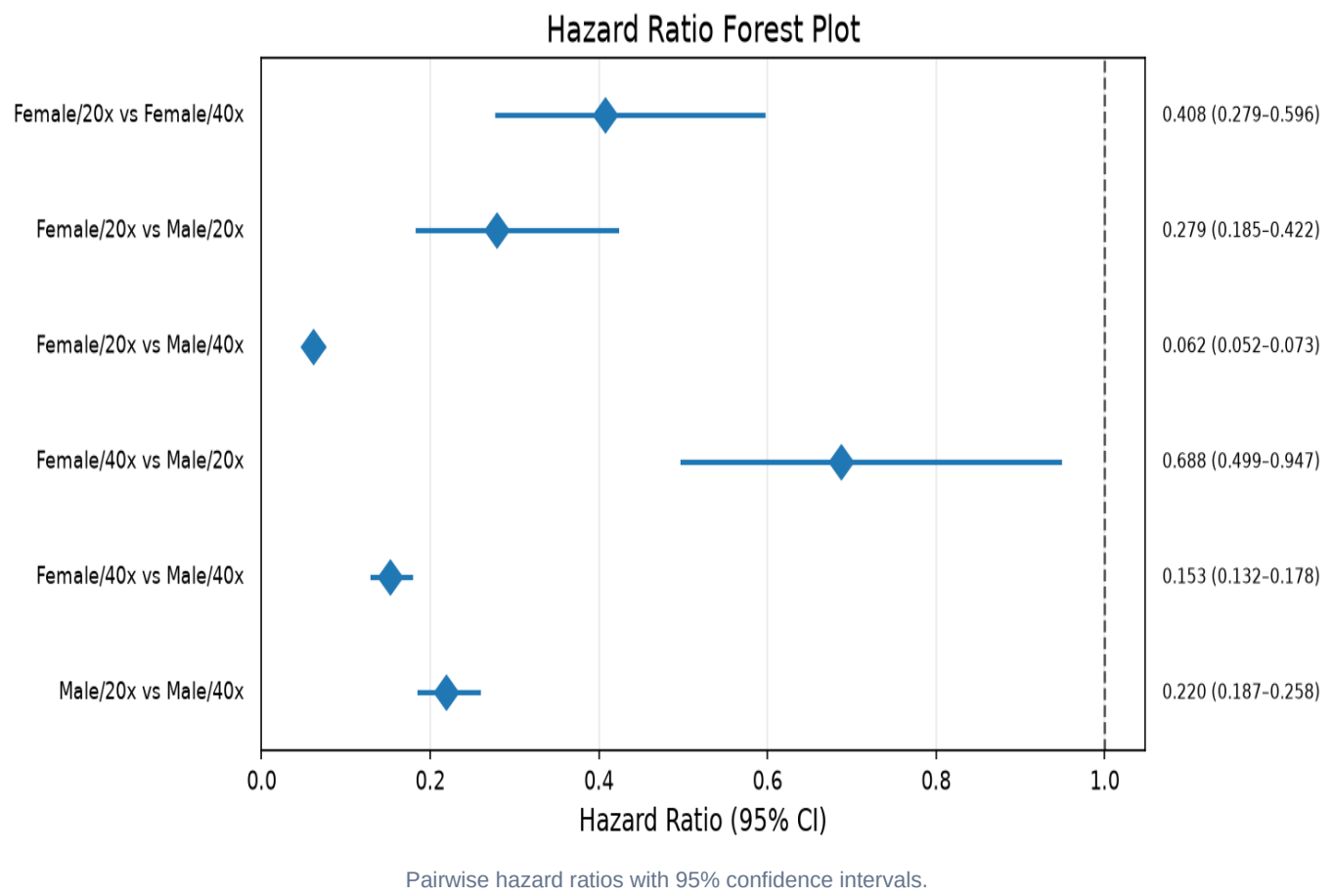
Nelson-Aalen cumulative hazard



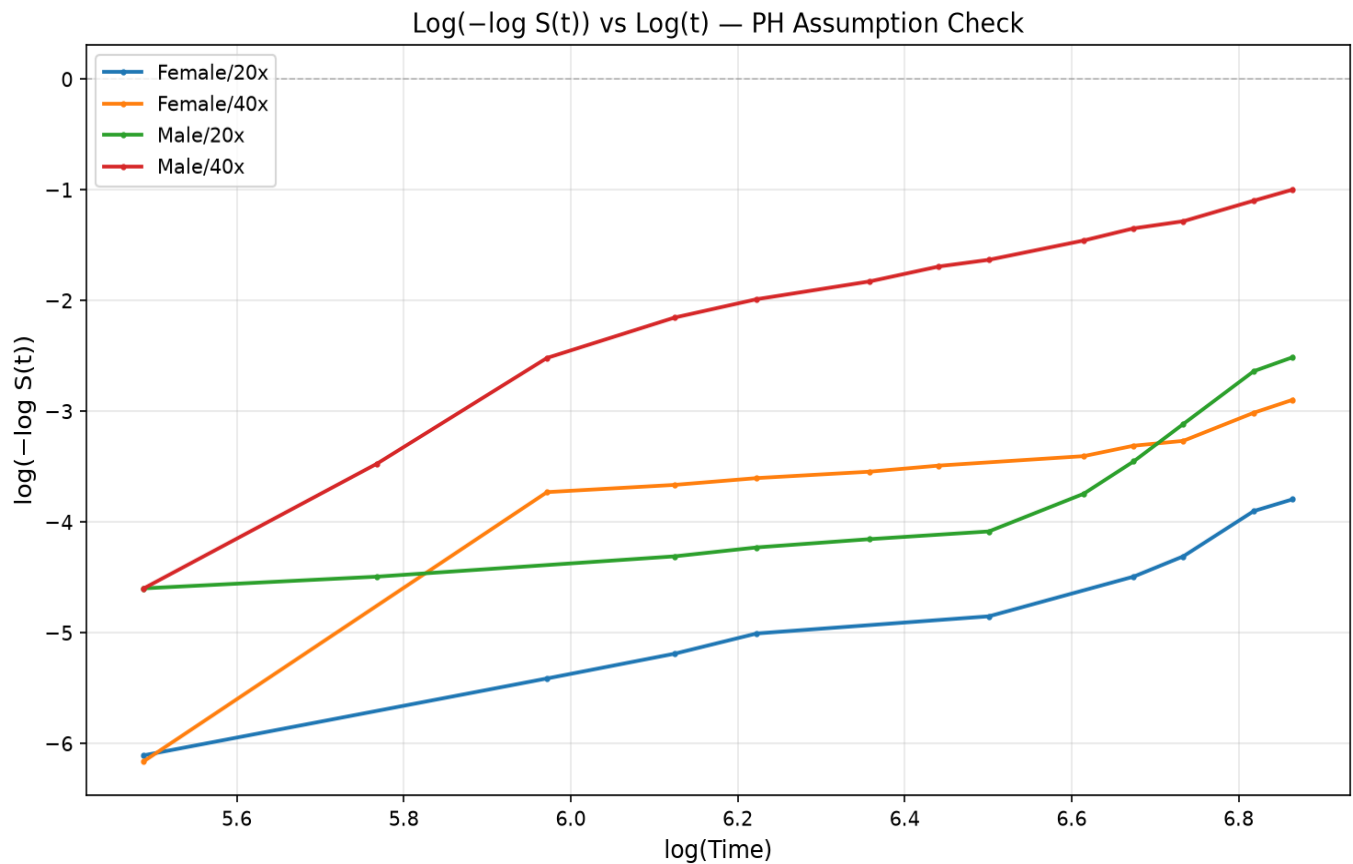
Number at risk



Hazard-ratio forest



Log-log diagnostic



Parallel lines support the proportional-hazards assumption.

Lifespan statistics

Median survival

treatment	median_survival
Female/20x	—
Female/40x	—
Male/20x	—
Male/40x	—

Mean survival

treatment	rmst	restriction_time
Female/20x	951.565	958.054
Female/40x	937.129	958.054
Male/20x	939.142	958.054
Male/40x	845.827	958.054

Lifespan statistics by treatment

group	n	n_deaths	n_censored	mean_rmst	median	t_max
Female/20x	900	20	880	951.570	—	958.050
Female/40x	1900	102	1798	937.130	—	958.050
Male/20x	900	70	830	939.140	—	958.050
Male/40x	1900	585	1315	845.830	—	958.050

Lifespan statistics by factor level

group	n	n_deaths	n_censored	mean_rmst	median	t_max
Sex=Female	2800	122	2678	941.870	—	958.050
Sex=Male	2800	655	2145	875.860	—	958.050
Density=20x	1800	90	1710	945.530	—	958.050
Density=40x	3800	687	3113	891.940	—	958.050

Survival quantiles

treatment	S=90%	S=75%	S=50%	S=25%	S=10%
Female/20x	—	—	—	—	—
Female/40x	—	—	—	—	—
Male/20x	—	—	—	—	—
Male/40x	456.929	914.349	—	—	—

Survival comparisons

Overall comparison

Test	chi²	df	p	
Omnibus log-rank	740.82	3	<0.0001	***

Pairwise log-rank

group1	group2	observed_1	expected_1	chi2	p_value	df	p_bonferroni	significant_0.05
Female/20x	Female/40x	20.000	39.606	14.491	0.000	1	0.001	True
Female/20x	Male/20x	20.000	45.513	29.176	0.000	1	0.000	True
Female/20x	Male/40x	20.000	215.050	281.304	0.000	1	0.000	True
Female/40x	Male/20x	102.000	116.851	5.943	0.015	1	0.089	False
Female/40x	Male/40x	102.000	365.958	416.488	0.000	1	0.000	True
Male/20x	Male/40x	70.000	231.014	178.010	0.000	1	0.000	True

Pairwise Gehan-Wilcoxon

group1	group2	chi2	p_value	df	p_bonferroni	significant_0.05
Female/20x	Female/40x	14.633	0.000	1	0.001	True
Female/20x	Male/20x	29.099	0.000	1	0.000	True
Female/20x	Male/40x	279.819	0.000	1	0.000	True
Female/40x	Male/20x	5.629	0.018	1	0.106	False
Female/40x	Male/40x	410.245	0.000	1	0.000	True
Male/20x	Male/40x	181.411	0.000	1	0.000	True

Pairwise hazard ratios

group1	group2	hazard_ratio	hr_ci_lo	hr_ci_hi
Female/20x	Female/40x	0.408	0.279	0.596
Female/20x	Male/20x	0.279	0.185	0.422
Female/20x	Male/40x	0.062	0.052	0.073
Female/40x	Male/20x	0.688	0.499	0.947
Female/40x	Male/40x	0.153	0.132	0.178
Male/20x	Male/40x	0.220	0.187	0.258

Data quality

No chambers were excluded from this analysis.

DataSet2

Standard Lifespan

Experiment summary

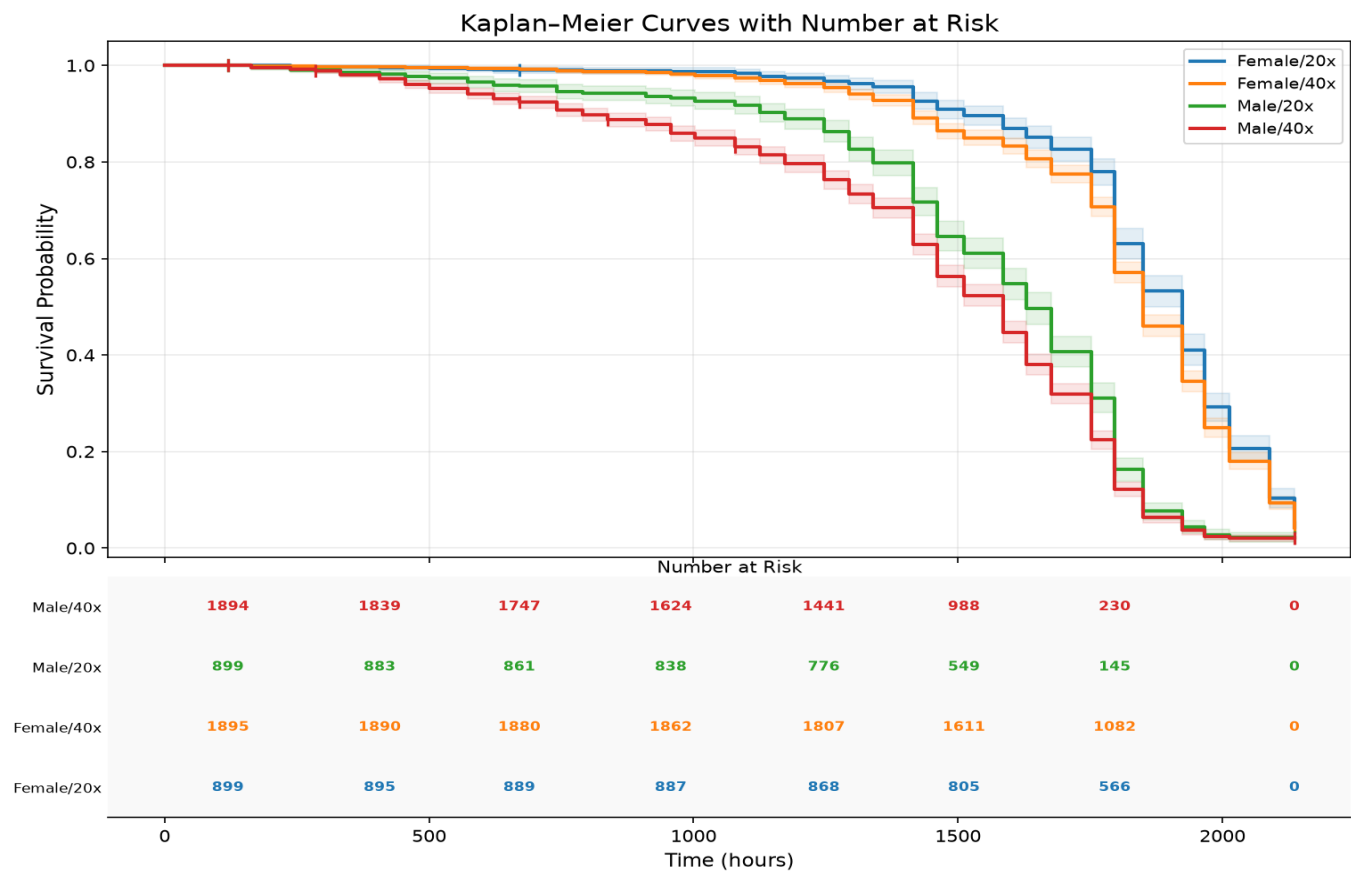
Per-treatment summary

treatment	n_individuals	n_deaths	n_censored	pct_censored
Female/20x	900	859	41	4.600
Female/40x	1900	1815	85	4.500
Male/20x	900	878	22	2.400
Male/40x	1900	1850	50	2.600

Observation window 119.62–2136.14, 200 chamber(s).

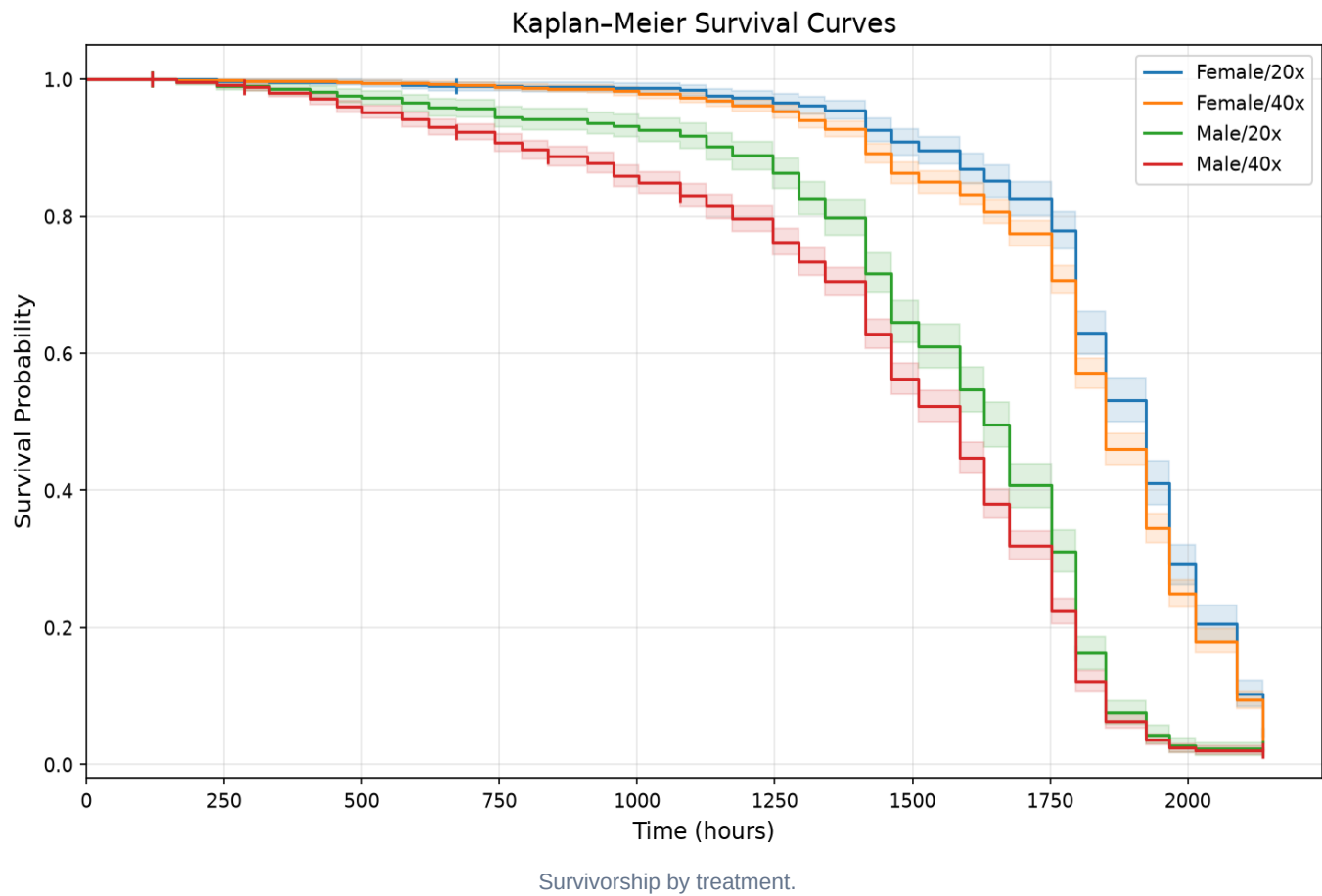
Survivorship figures

KM curves with at-risk table — headline figure

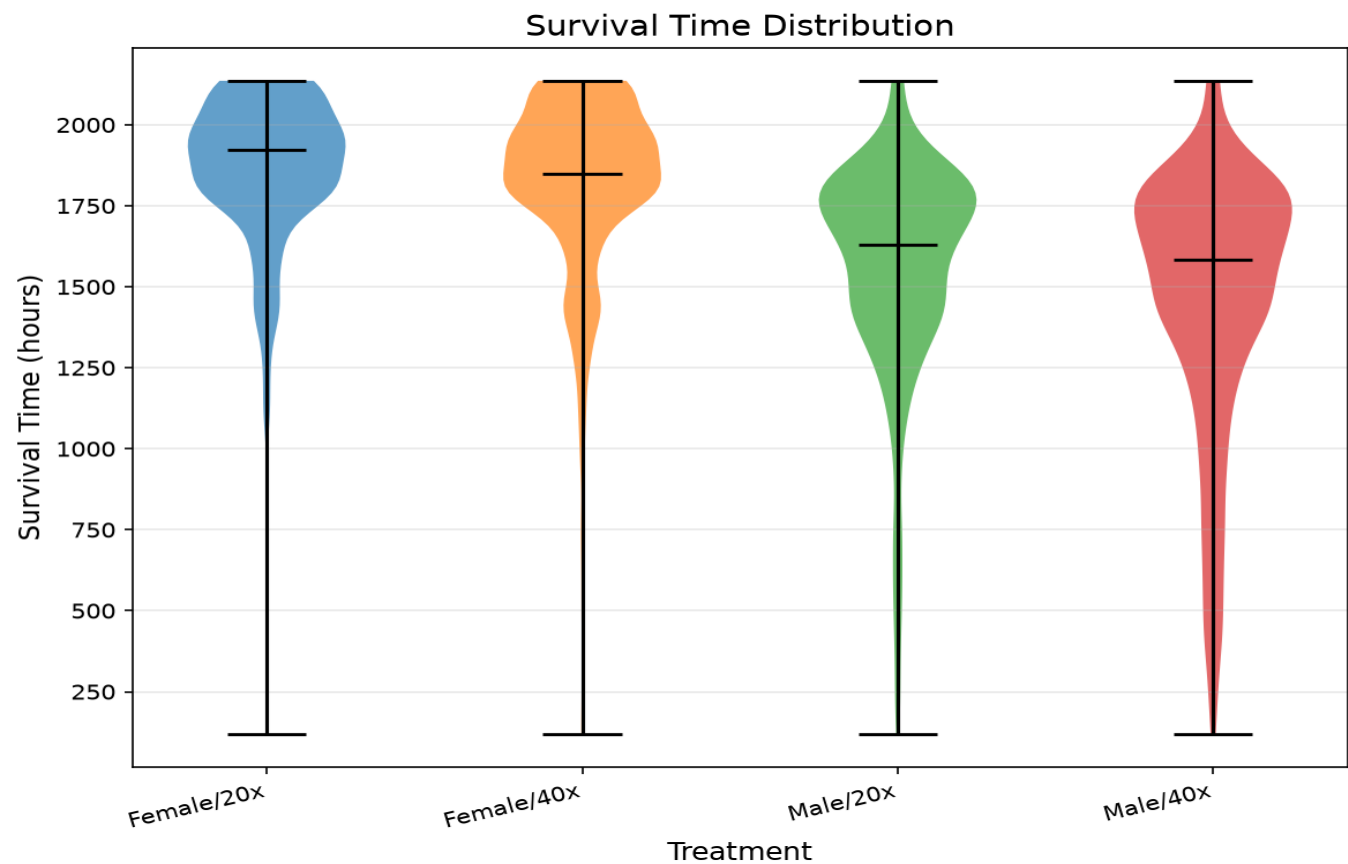


Survivorship with the number at risk beneath the axis.

Kaplan-Meier curves

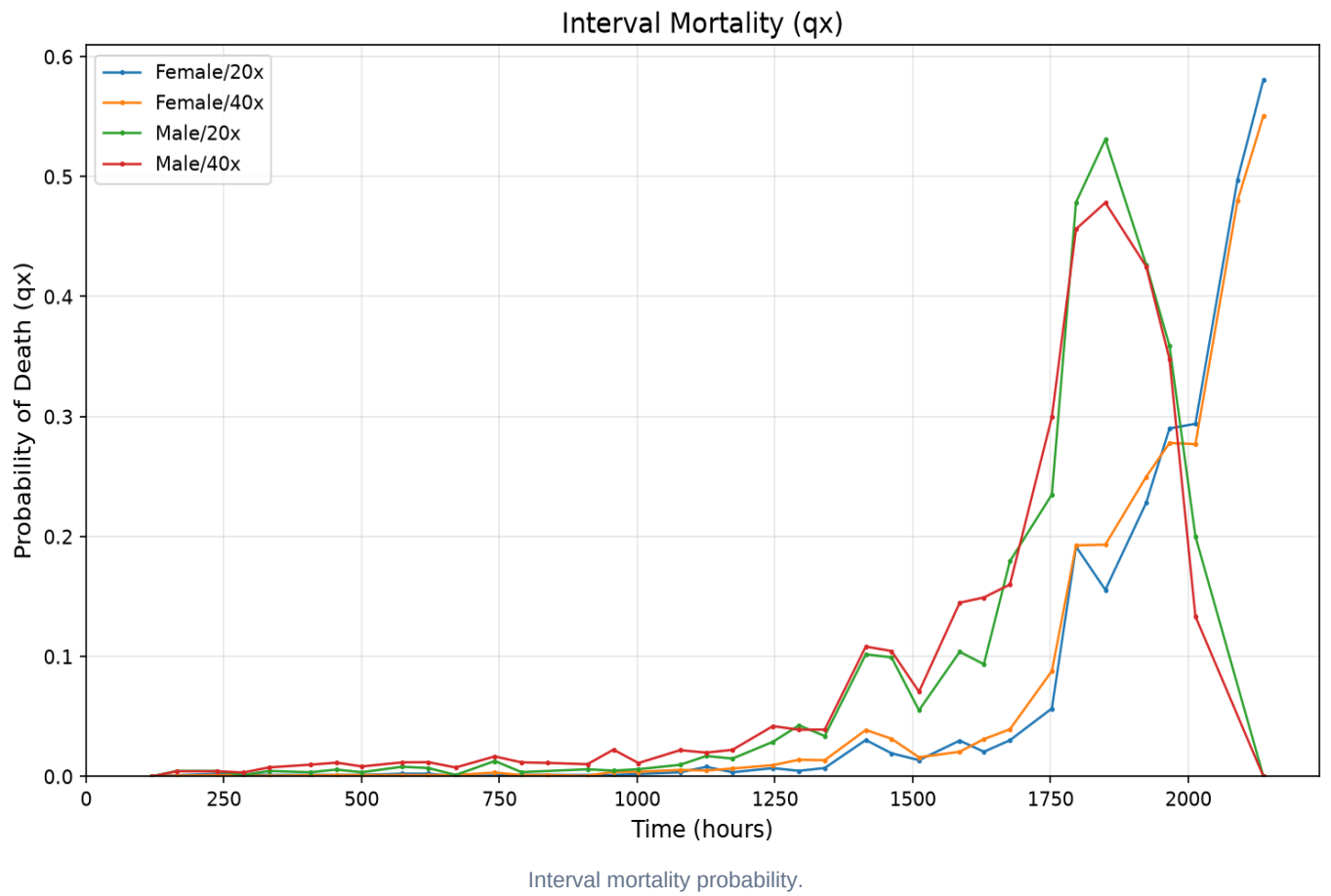


Lifespan distribution

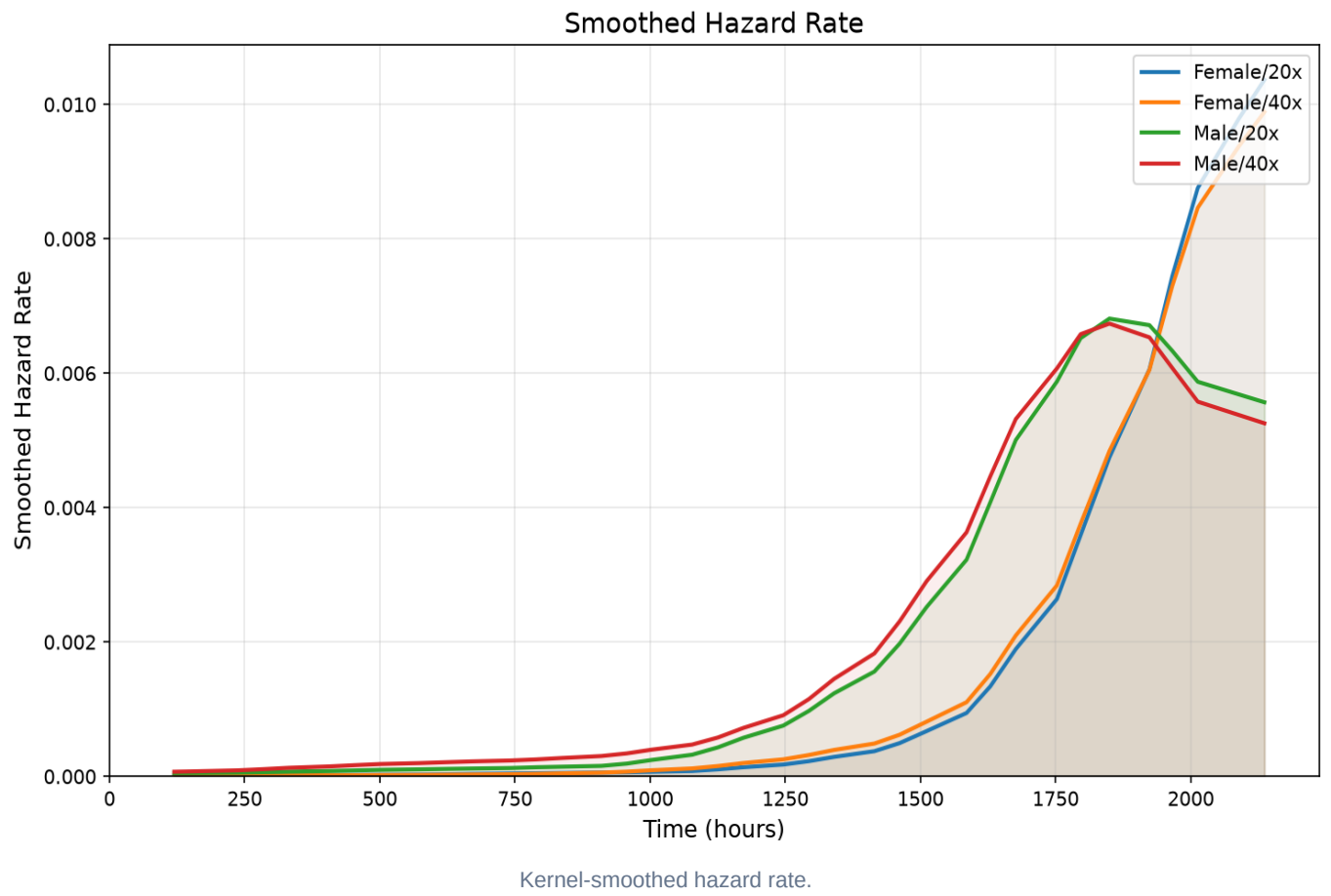


Distribution of individual lifespans by treatment.

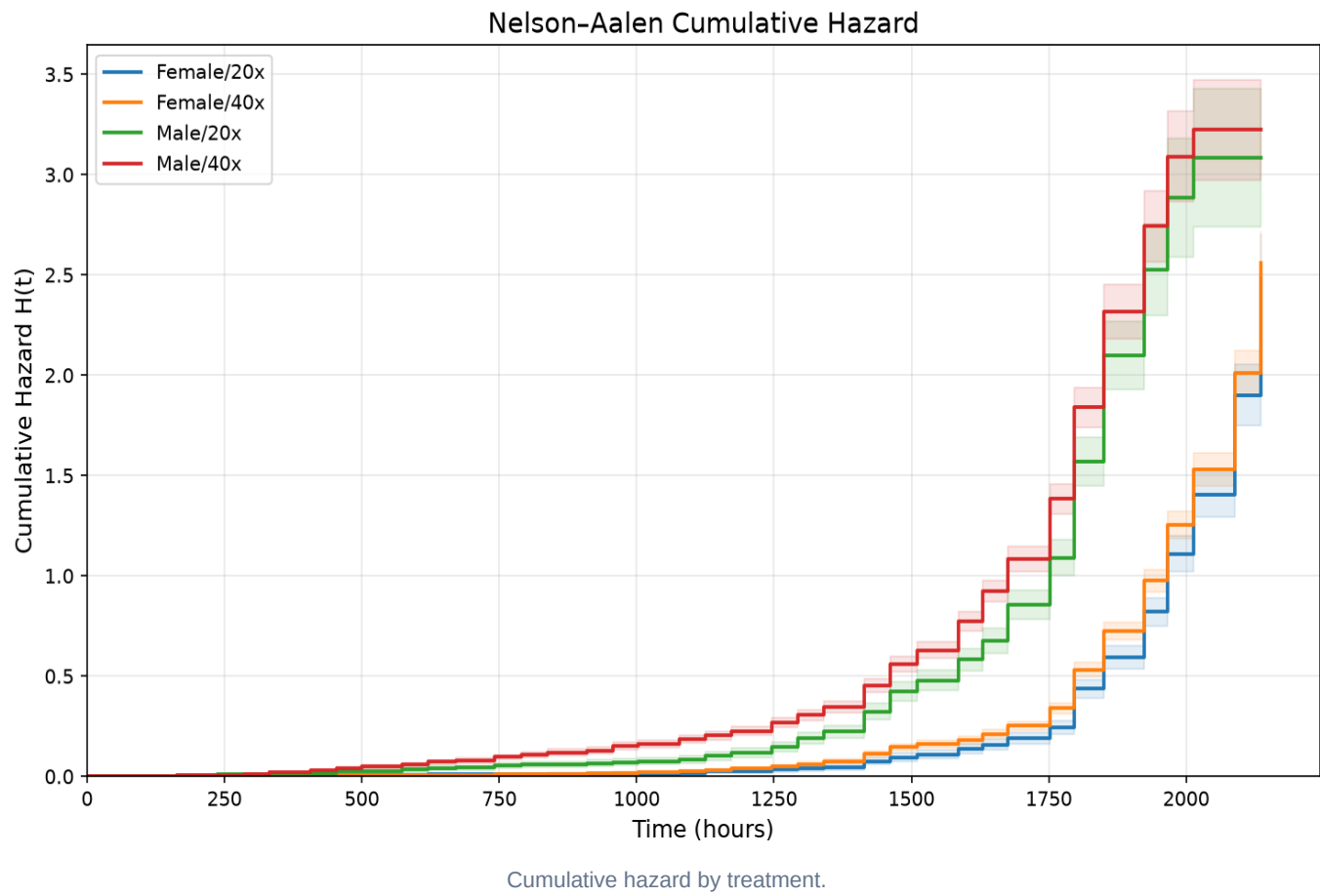
Mortality (qx)



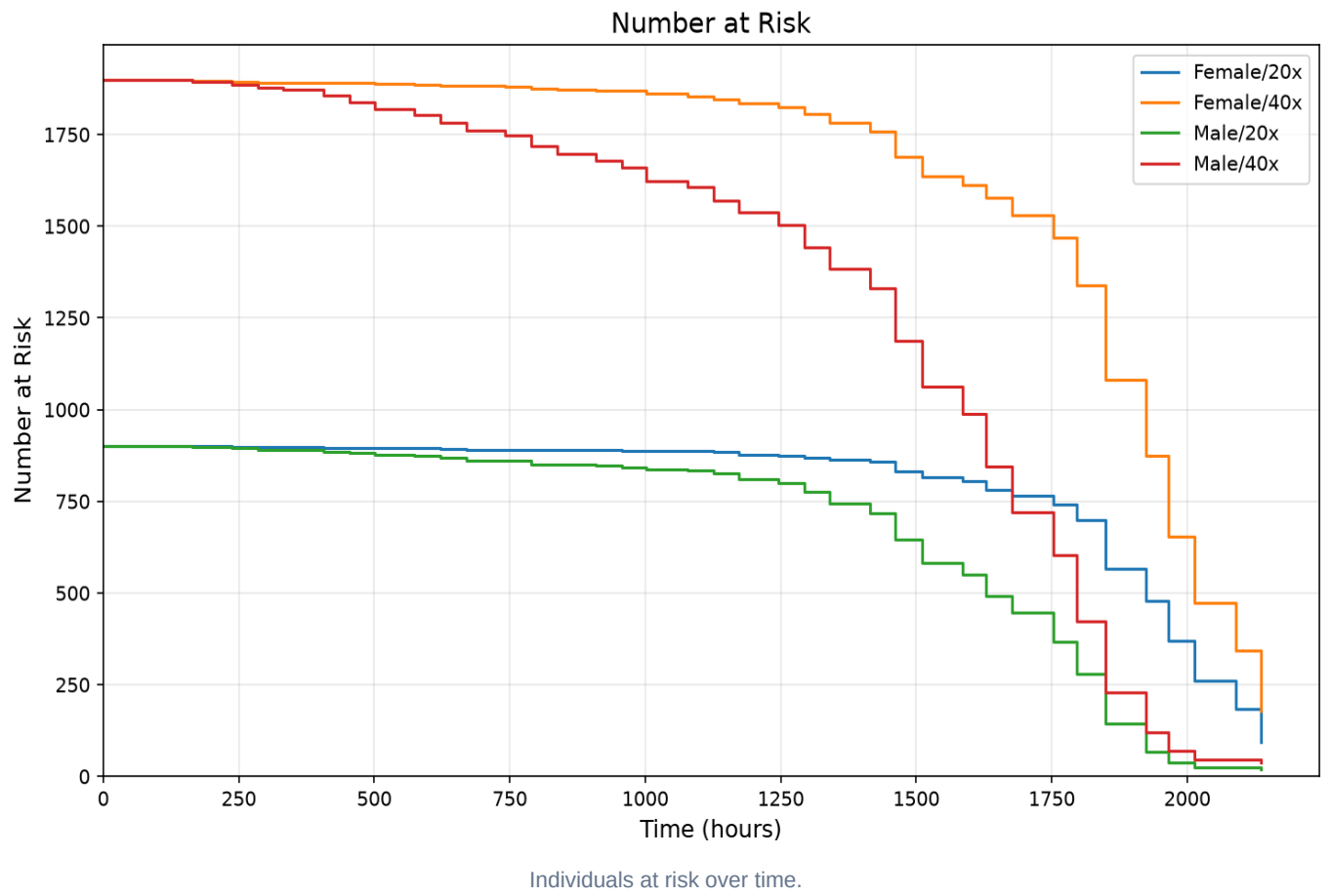
Smoothed hazard



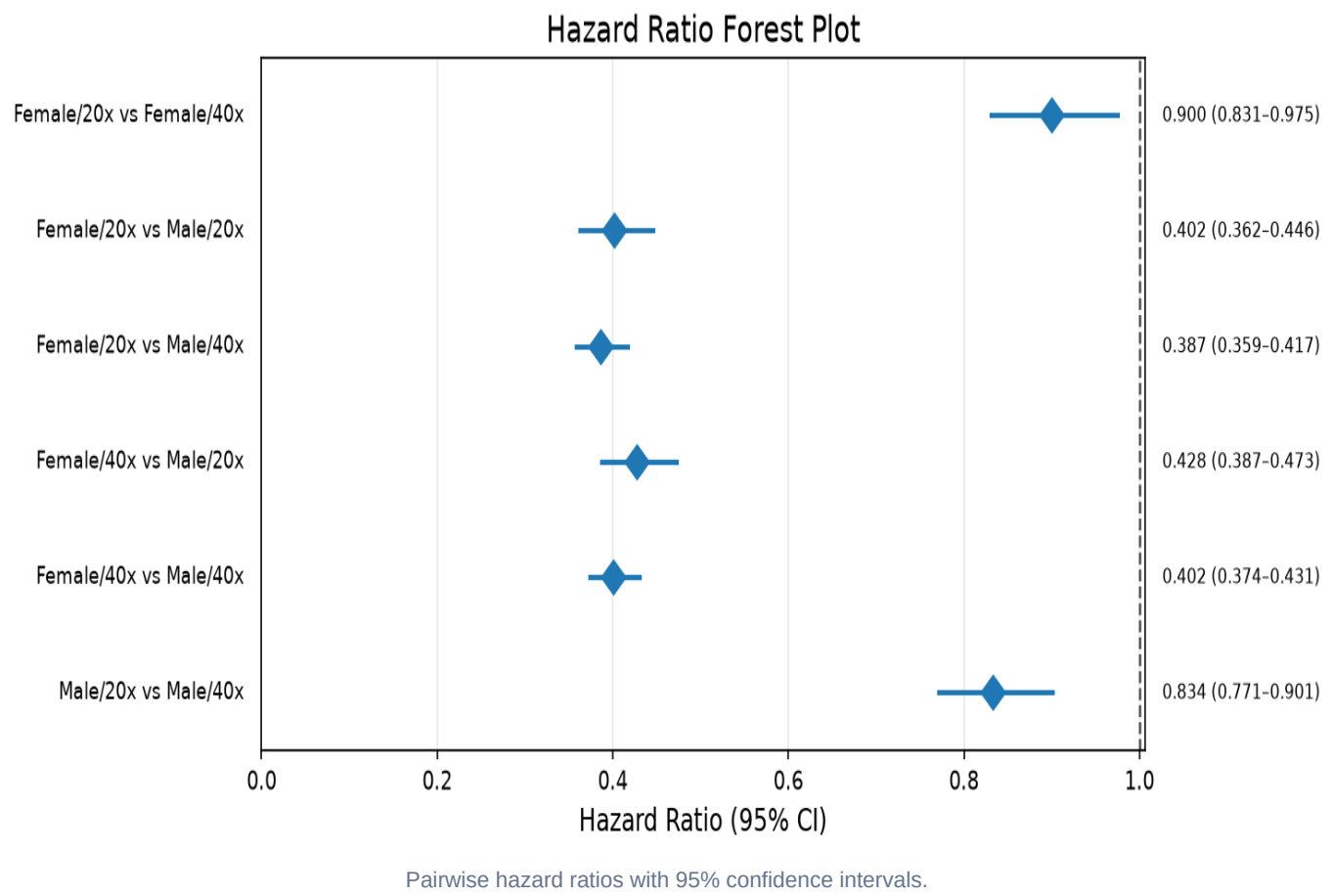
Nelson-Aalen cumulative hazard



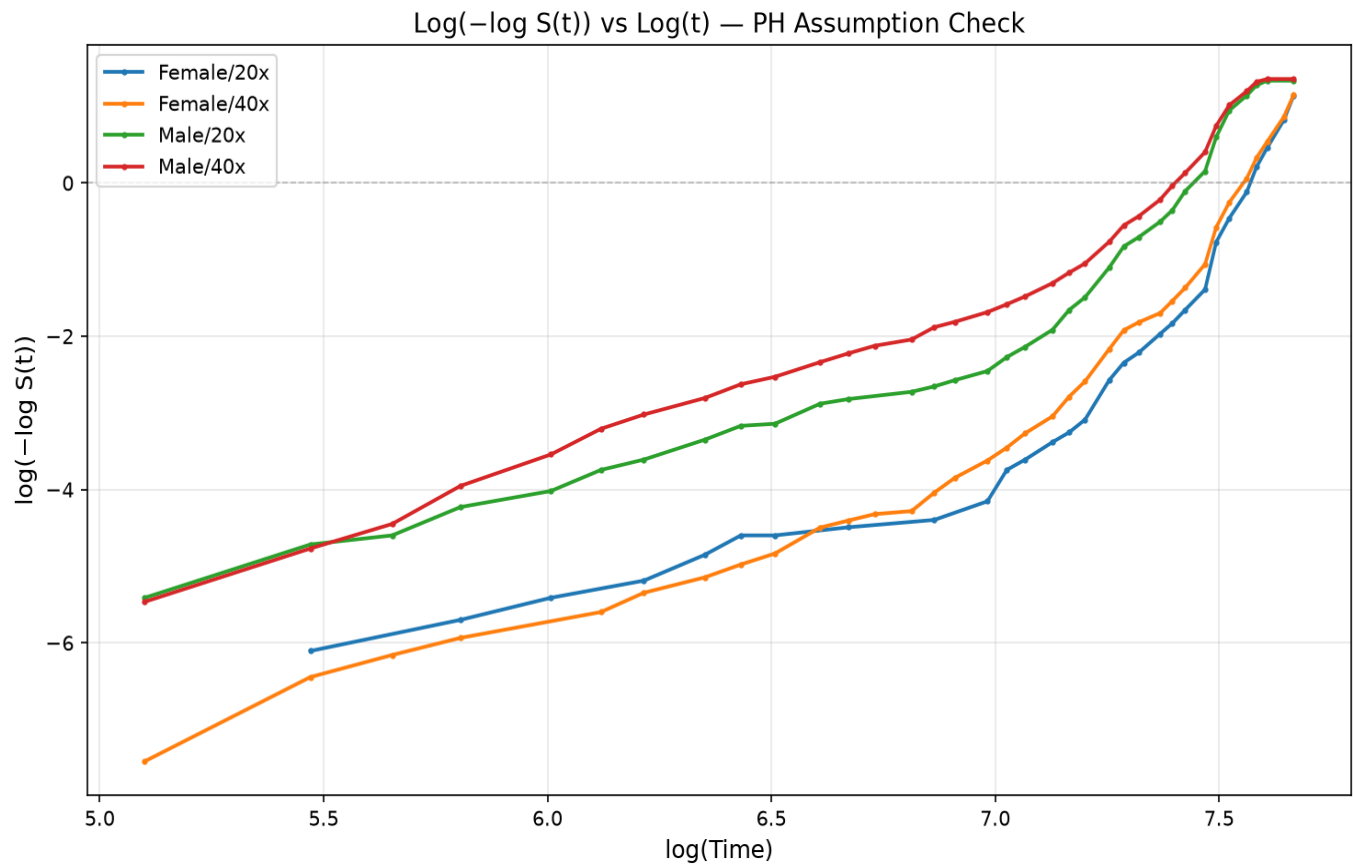
Number at risk



Hazard-ratio forest



Log-log diagnostic



Parallel lines support the proportional-hazards assumption.

Lifespan statistics

Median survival

treatment	median_survival
Female/20x	1923.479
Female/40x	1849.431
Male/20x	1628.977
Male/40x	1585.109

Mean survival

treatment	rmst	restriction_time
Female/20x	1825.136	2136.138
Female/40x	1785.516	2136.138
Male/20x	1529.745	2136.138
Male/40x	1431.423	2136.138

Lifespan statistics by treatment

group	n	n_deaths	n_censored	mean_rmst	median	t_max
Female/20x	900	859	41	1825.140	1923.480	2136.140
Female/40x	1900	1815	85	1785.520	1849.430	2136.140
Male/20x	900	878	22	1529.740	1628.980	2136.140
Male/40x	1900	1850	50	1431.420	1585.110	2136.140

Lifespan statistics by factor level

group	n	n_deaths	n_censored	mean_rmst	median	t_max
Sex=Female	2800	2674	126	1798.370	1849.430	2136.140
Sex=Male	2800	2728	72	1463.150	1585.110	2136.140
Density=20x	1800	1737	63	1677.540	1796.280	2136.140
Density=40x	3800	3665	135	1608.720	1752.370	2136.140

Survival quantiles

treatment	S=90%	S=75%	S=50%	S=25%	S=10%
Female/20x	1511.224	1796.282	1923.479	2012.853	2136.138
Female/40x	1414.742	1752.370	1849.431	1965.732	2089.093
Male/20x	1172.736	1414.742	1628.977	1796.282	1849.431
Male/40x	789.977	1293.467	1585.109	1752.370	1849.431

Survival comparisons

Overall comparison

Test	chi²	df	p	
Omnibus log-rank	1554.87	3	<0.0001	***

Pairwise log-rank

group1	group2	observed_1	expected_1	chi2	p_value	df	p_bonferroni	significant_0.05
Female/20x	Female/40x	859.000	921.307	8.241	0.004	1	0.025	True
Female/20x	Male/20x	859.000	1231.094	498.050	0.000	1	0.000	True
Female/20x	Male/40x	859.000	1477.824	752.562	0.000	1	0.000	True
Female/40x	Male/20x	1815.000	2230.979	554.198	0.000	1	0.000	True
Female/40x	Male/40x	1815.000	2600.681	1025.915	0.000	1	0.000	True
Male/20x	Male/40x	878.000	989.550	24.441	0.000	1	0.000	True

Pairwise Gehan-Wilcoxon

group1	group2	chi2	p_value	df	p_bonferroni	significant_0.05
Female/20x	Female/40x	14.331	0.000	1	0.001	True
Female/20x	Male/20x	523.649	0.000	1	0.000	True
Female/20x	Male/40x	764.381	0.000	1	0.000	True
Female/40x	Male/20x	585.387	0.000	1	0.000	True
Female/40x	Male/40x	1082.786	0.000	1	0.000	True
Male/20x	Male/40x	36.552	0.000	1	0.000	True

Pairwise hazard ratios

group1	group2	hazard_ratio	hr_ci_lo	hr_ci_hi
Female/20x	Female/40x	0.900	0.831	0.975
Female/20x	Male/20x	0.402	0.362	0.446
Female/20x	Male/40x	0.387	0.359	0.417
Female/40x	Male/20x	0.428	0.387	0.473
Female/40x	Male/40x	0.402	0.374	0.431
Male/20x	Male/40x	0.834	0.771	0.901

Data quality

No chambers were excluded from this analysis.